TASK 3 FINAL SUBMISSION DOCUMENT

1. TITLE PAGE

Website Traffic & User Interaction Analysis – Data Analyst Internship

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Internship: Alfido Tech
Task: Task 3 – Website Traffic Analysis

2. EXECUTIVE SUMMARY

The current project aims at conducting exploratory analysis of website traffic and user interaction data to reveal patterns of website engagements, traffic patterns, and user behaviors. The dataset consists of interaction-related variables including events, countries, cities, artists, tracks, and links accessed.

The primary purpose of the analysis is to reveal patterns of user behavior, calculate engagement metrics, and make recommendations aimed at boosting website engagements and conversions for Alfido Tech.

To conduct EDA and visualize traffic and user interactions, Python libraries Pandas, Matplotlib, and Seaborn were used.

The key approaches adopted during the analysis include:

- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Traffic and interaction analysis
- User activity and trends visualization

The findings of the analysis show that some countries and cities contribute more traffic activity than others. Also, only several links account for most of the interactions. Popular artists and tracks tend to be visited frequently. Bounce behavior can also be found in several user sessions.

3. DATASET OVERVIEW

- Event
- Date
- Country
- City
- Artist

- Album
- Track
- ISRC
- Link ID

4. CODE & GITHUB LINK

GitHub Repository:

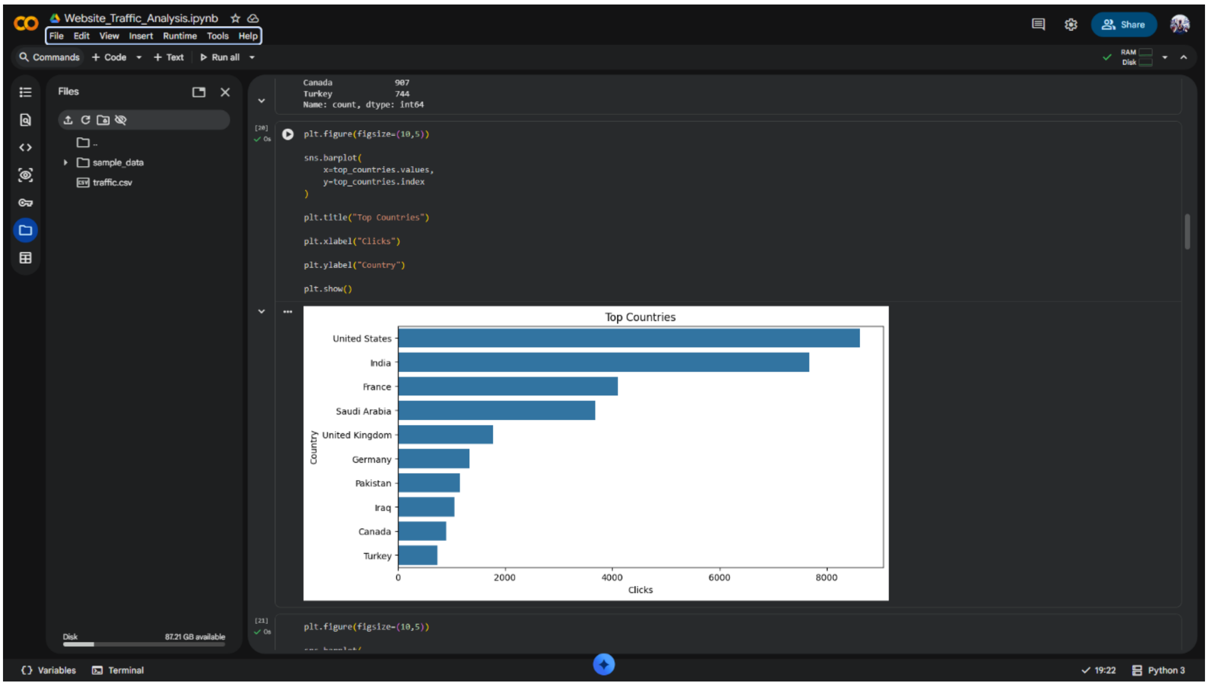
https://github.com/krishnaM04/Website-Traffic-Analysis.git

Notebook contains:

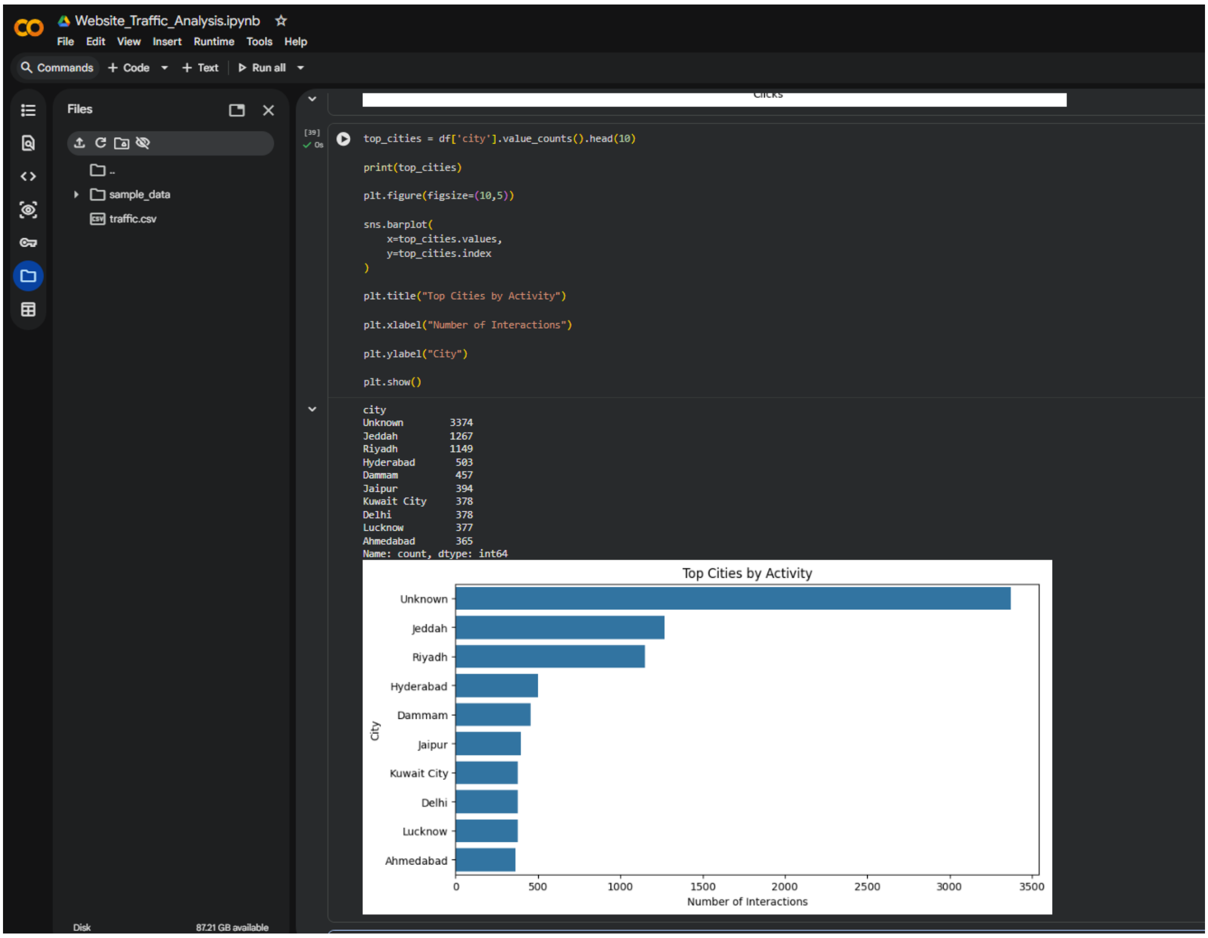
- Data cleaning
- Exploratory Data Analysis
- Traffic analysis
- Bounce rate estimation
- Visualizations

5. VISUALIZATIONS (ADD SCREENSHOTS HERE)

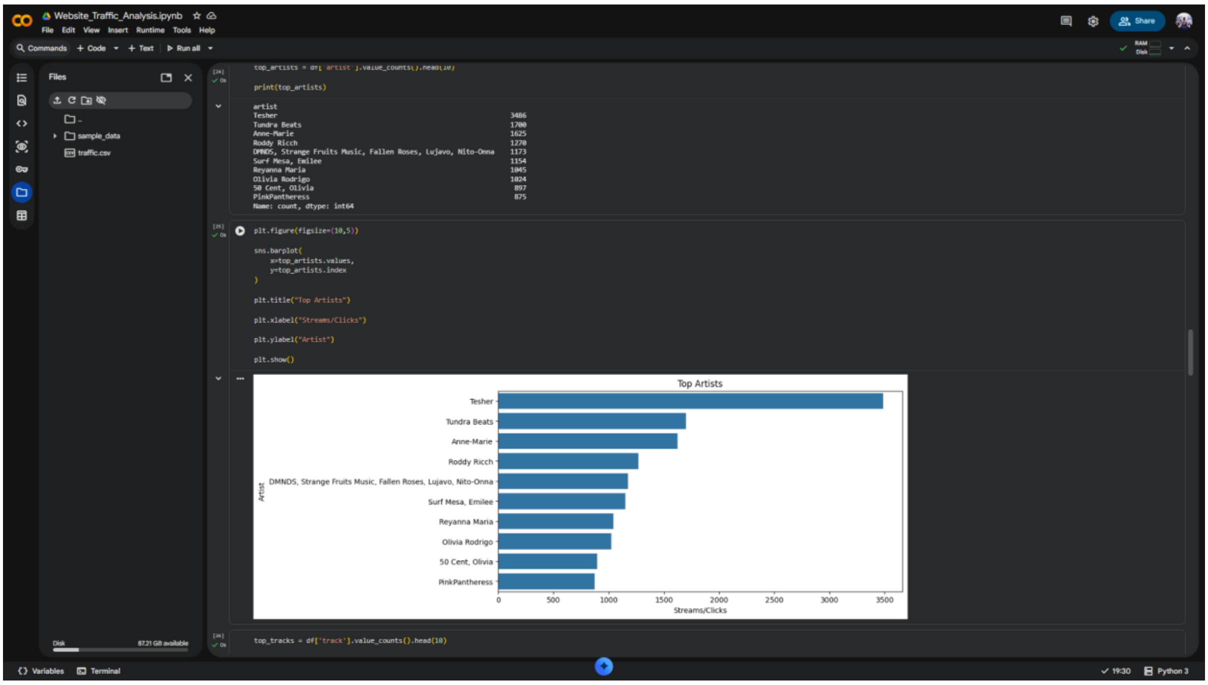
Insert screenshots of:



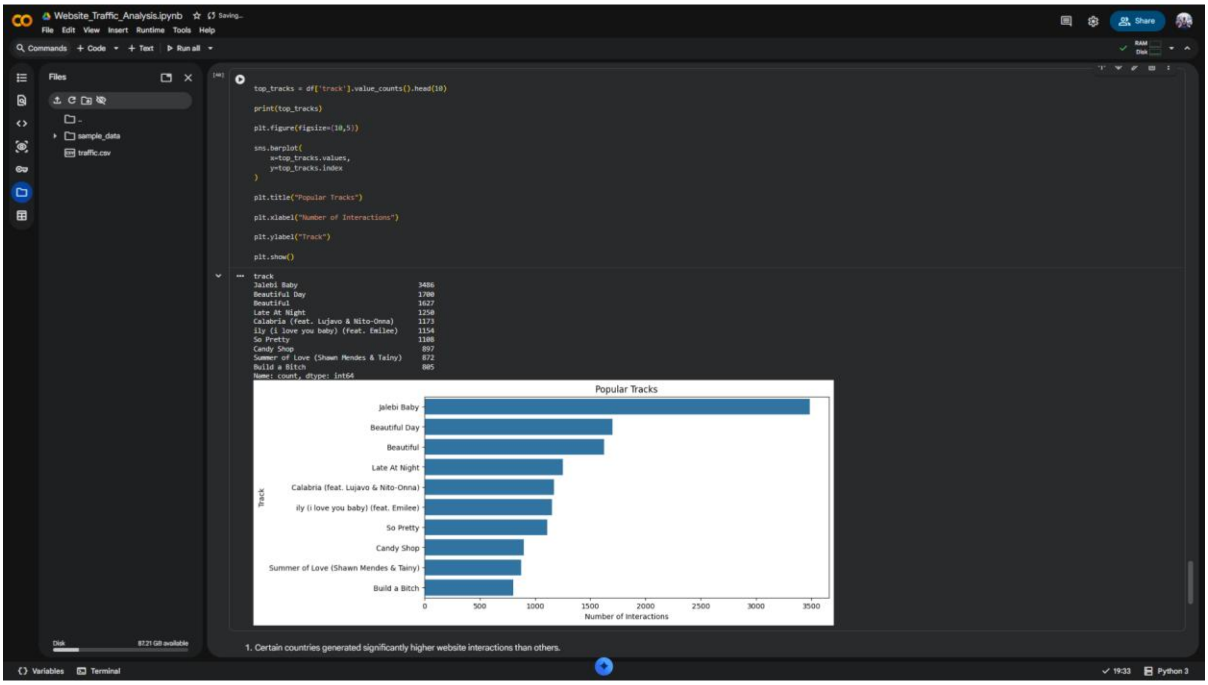
Top Countries by Traffic



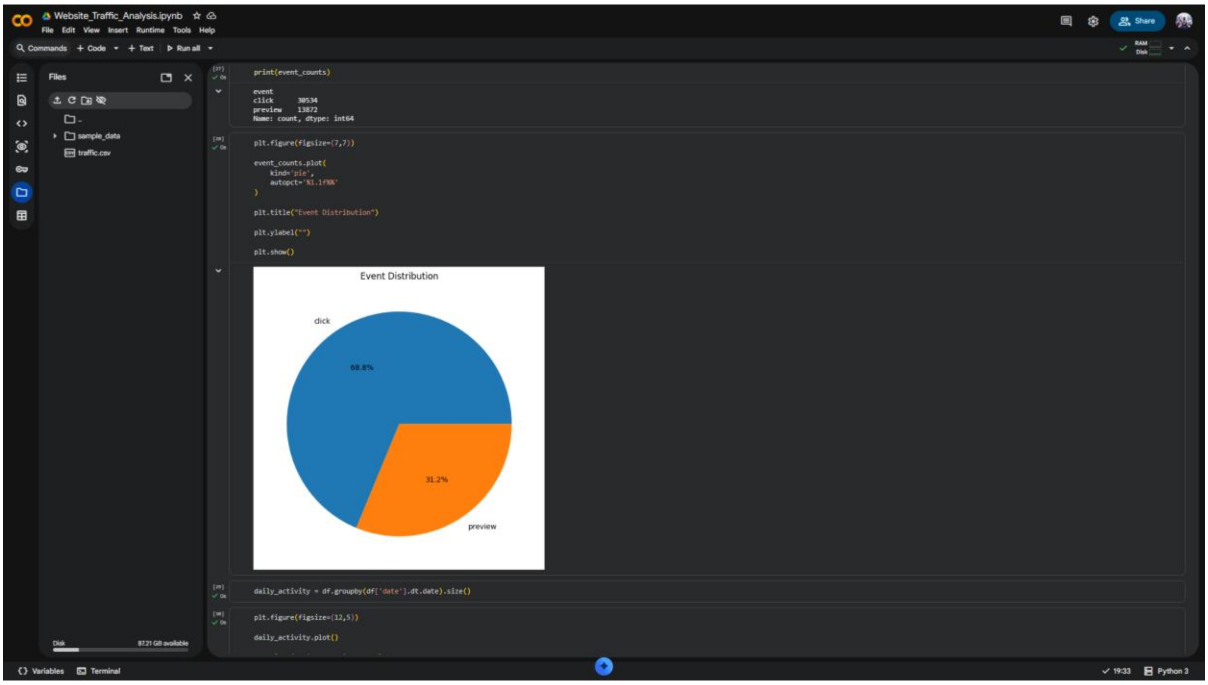
Top Cities by Activity



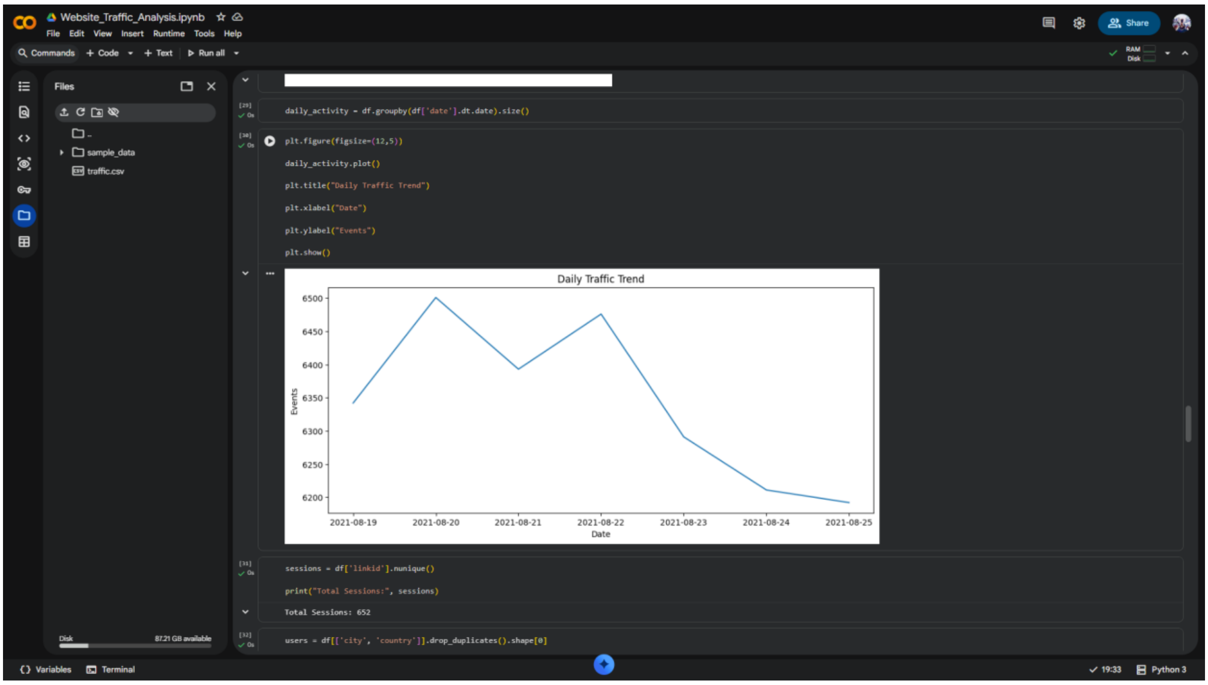
Popular Artists



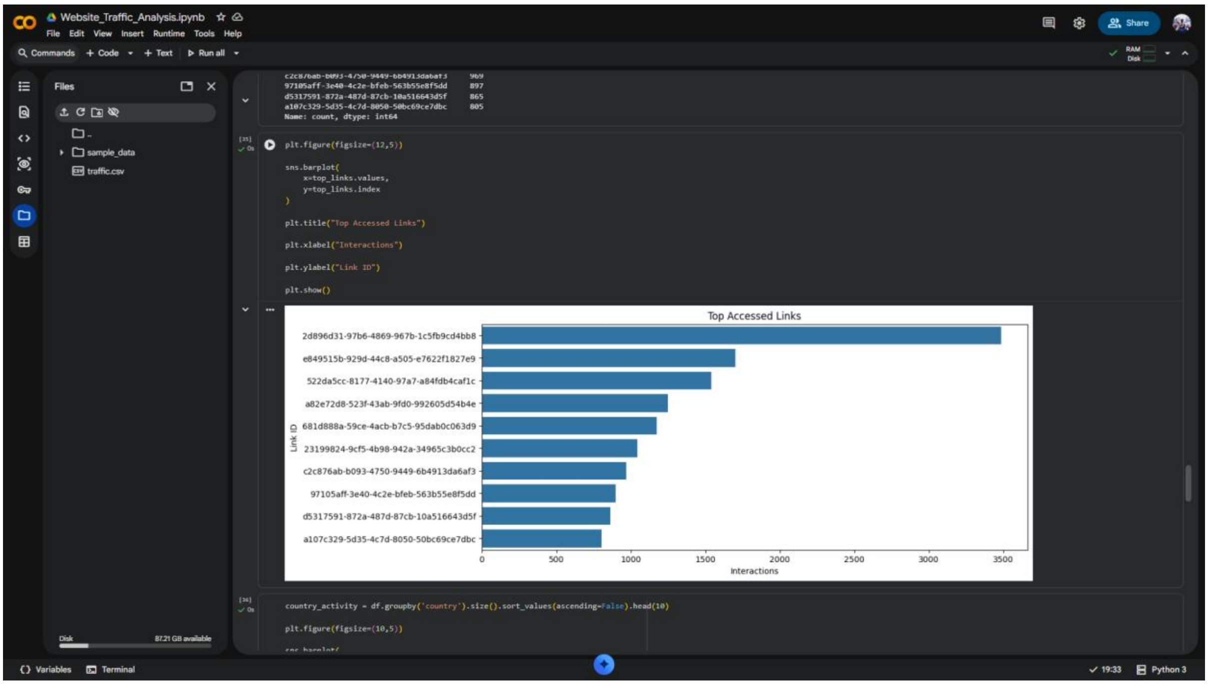
Popular Tracks



Event Distribution



Daily Traffic Trend



Top Accessed Links

6. KEY FINDINGS

- Certain countries generated significantly higher interaction activity
- A few links received the majority of user interactions
- Popular artists and tracks attracted higher engagement
- Traffic activity varied across different dates
- Bounce behavior was observed in multiple sessions

7. TRAFFIC & ENGAGEMENT METRICS

- Estimated Sessions
- Estimated Users
- Bounce Rate
- Link Interaction Analysis
- Traffic Trend Analysis

Bounce Rate Formula:

$$\text{Bounce Rate} = \frac{\text{Single Interaction Sessions}}{\text{Total Sessions}} \times 100$$

8. RECOMMENDATIONS

1. Improve engagement on low-performing links
2. Focus marketing campaigns on high-traffic regions
3. Promote highly engaging artists and tracks
4. Reduce bounce behavior through better navigation
5. Analyze peak traffic periods for marketing optimization

9. DATASET LIMITATION NOTE

The dataset did not contain traditional website analytics fields such as session IDs, user IDs, landing pages, exit pages, or session duration. Therefore, sessions, users, and bounce rates were estimated using available interaction and link data.

10. CONCLUSION

In the current project, website traffic and user interaction data have been analyzed in order to explore user engagement, traffic activity, and content popularity. In this regard, all calculations and visualizations were performed with the help of such Python libraries as Pandas, Matplotlib, and Seaborn.

It is possible to mention numerous valuable insights provided by the analysis conducted in the current project. In particular, it was found out that there are differences in user activities in terms of geographical locations, with some countries and cities demonstrating higher traffic and engagement level than others. Furthermore, popular artists, songs, and websites generated higher traffic and interaction rate. The traffic trend analysis showed that there are different engagement levels on different dates.

Despite the fact that the current dataset does not include standard website analytics fields such as session ID, user ID, landing page, exit page, and session duration, other approaches could be applied in order to estimate sessions, users, and bounce rate. Various visualizations helped better analyze traffic and engagement trends.

Finally, based on the results obtained, several recommendations were developed.